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GPU Programming Specialization

- ✓

Reading: Course Overview
10 min
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Reading: Course Outline
10 min
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Video: GPU Programming Specialization
3 min
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Discussion Prompt: Large Scale Data and Challenges Discussion
15 min

Course Technical and Programming Expectations

Coursera Lab and Assignment Usage

Course Outline

Modules Overview

Module 1 - Course Overview

- *Lessons:* GPU Programming Specialization, Course Technical and Programming Expectations, Coursera Lab and Assignment Usage,
- *Assignment:* Simple CUDA Programming assignment

Module 2 - Threads, Blocks and Grids

- *Lessons:* CUDA Kernels, Converting CPU Divide and Conquer to Parallel Kernel Execution, Defining Kernel Thread and Data Layout via Blocks, 3-Dimensional Thread and Data Layout using Grids and Blocks, 3-Dimensional Thread and Data Layout using Grids and Blocks, Execution of CUDA Kernel in 3-Dimensional Data/Thread Space
- *Assignments:* Data Search, Performing RGB to Grayscale on Image Data
- Quizzes: CPU to GPU Algorithm Conversion, Multidimensional Data and Computation on the GPU

Module 3 - Host and Global Memory

- *Lessons:* Nvidia GPU Global Device Memory, Nvidia GPU Global Memory Analysis, Host Memory Allocation, Device Memory Allocation
- *Assignments:* Allocation and Assignment of Different Types of Host and Global Memory
- Quizzes: CPU and GPU Global Memory

Module 4 - Shared and Constant Memory

- Lessons: Nvidia Shared/Constant Memory Hardware, Nvidia GPU Shared and Constant Memory Analysis, Nvidia Shared Memory Allocation, Nvidia Constant Memory Allocation
- Assignments: CUDA Shared and Constant Memory Image Manipulation
- Quiz: CUDA Constant and Shared Memory

Module 5 - Register Memory

- Lessons: Nvidia GPU Register Memory, Nvidia GPU Register Memory Analysis, Register Memory Allocation, Nvidia Memory Comparison
- Assignments: CUDA Device Memory Analysis
- Quiz: Device Memory

Grading:

Assignments - This is 75% of total grade. You need a score of 80% to pass the assignments. There are 8 assignments that are weighted in the following way:

header	% of Assignment Grade
Simple CUDA Assignment	10
Data Search Assignment	10
Performing RGB to Grayscale on Image Data	20
Allocation and Assignment of Different Types of Host and Global Memory	20
CUDA Shared and Constant Memory Image Manipulation	20
CUDA Device Memory Analysis	20

There are no assignments dropped.

Quizzes - This is 25% of total grade. You need 80% to pass the quiz section. There are 4 quizzes that are weighted equally, so 25% of quizzes grade. There are no dropped quizzes.

You will need to pass both assignments and quizzes sections to pass the course.

Go to next item ✓ Completed

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